Enhanced Bang-Bang Control Method for DC Voltage Regulation in Non-Minimum Phase Power Electronics Converter

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***Abstract*—The bang-bang (BB) control strategy, recognized for its simplicity and rapid dynamic response, is widely adopted in power electronic converters. However, conventional BB methods suffer from high current ripple due to the one-sample delay introduced by digital implementation. To address this limitation, this paper proposes a model-based bang-bang (MBB) control method for DC-DC boost converters. The proposed approach utilizes a converter model to update the state variables in real time and compensate the one-sample delay inherent in digital control, thereby mitigating its adverse effects on current ripple and voltage regulation. Furthermore, a corrective current term is incorporated into the BB switching function to eliminate steady-state error and enhance dynamic performance. To improve practicality, an output-current estimation technique is developed to reduce dependence on load characteristics and remove the need for an additional current sensor. Simulation and experimental results verify that the proposed MBB control retains the simplicity of BB control while significantly reducing current ripple and steady-state error, offering an efficient and easily implementable solution for high-performance DC-DC converters.**

***Index Terms*—Bang-bang control, boost converter, current regulation, digital implementation, model-based control, voltage regulation, delay compensation.**

I. INTRODUCTION

DC Microgrids offer higher efficiency than conventional AC systems by reducing multiple AC–DC conversions [1], [2]. Power electronic converters enable integration of renewable sources such as solar, wind, and batteries. Among them, DC–DC boost converters (Fig. 1) are widely used for voltage step-up and power balancing in DC microgrids [3]–[5]. However, their non-minimum-phase (NMP) nature causes the output voltage to initially move opposite to the control action, requiring advanced control strategies to ensure stability and accurate voltage regulation [6].

The most widely used approach to mitigate the NMP issue is the cascaded control structure, combining an inner current loop and an outer voltage loop. The inner loop can be implemented using different types of current controllers, such as PI [7], hysteresis [8], bang-bang (BB) [9], or model predictive control (MPC) [10], [11], while the outer loop typically uses a PI voltage regulator. This structure provides a clear reference current, simplifying current limitation and protection, but the outer PI voltage loop slows the transient response compared with faster model-based or MPC methods. In addition to classic cascaded structure, numerous advanced control strategies have been also reported in the literature [12], [13] to directly control the output voltage, including adaptive PID [14], sliding mode [15], and direct voltage model predictive control [16] methods. While effective, these methods are often limited by converter nonlinearities and digital processing delays [17]. In particular, in boost converters, an increase in duty cycle first causes a voltage drop before rising, and controller structures address this differently: PI controllers need a careful tuning, MPC needs a long-horizon prediction, and fast switching schemes like hysteresis, BB, or short-horizon MPC struggle to compensate for the NMP effect (Fig. 2(a)). In the case of BB control, the switching action determined solely by the voltage error can cause a temporary short circuit of the input source and uncontrolled inductor-current growth (Fig. 2(b)). This discussion defines the core problem analyzed in this paper: although BB control offers fast dynamic response, it leads to instability when applied directly to NMP systems such as the boost converter, whereas cascaded control ensures stability at the cost of slower dynamics and higher overshoot. Accordingly, this work aims to develop a BB-based voltage controller that achieves stability and fast response under both reference and load variations.

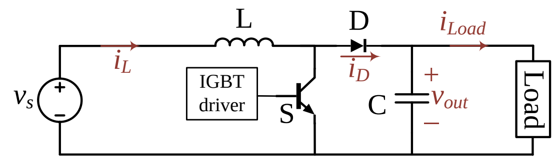

Fig. 1. DC-DC boost converter scheme.

The hysteresis control method is a widely used technique in power electronics [18], [19]. This nonlinear technique operates by defining a hysteresis band within which the system remains stable. When the error exceeds the hysteresis band, the system switches states to maintain control. However, this

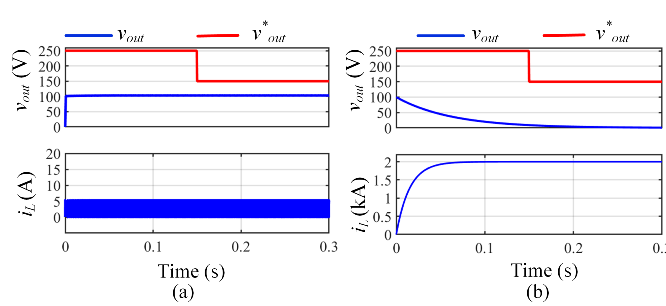

Fig. 2. Direct voltage regulation without considering current feedback using (a) single-horizon MPC, and (b) BB control.

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method causes variable switching frequency and current ripple, requiring careful design [20]. The BB control method, also known as on-off control, is a specific form of hysteresis control with a zero-width hysteresis band [21]. Unlike standard hysteresis control, it switches directly between maximum and minimum input levels whenever the error changes sign. This abrupt action ensures fast dynamic response, making BB control attractive for applications requiring rapid correction, such as motor drives and power converters [21]–[23]. Despite its simplicity, conventional BB control suffers from high current ripple and strong sensitivity to digital delays. In particular, a one-sample delay in digital controllers can distort switching behavior, increasing ripple and steady-state error [24]. Addressing this delay is therefore essential for stable and accurate implementation [25].

Model-based control strategies have gained significant attention for leveraging system dynamics to improve performance, enhancing transient response, robustness, and stability over conventional linear methods. They are particularly effective for nonlinear systems, where linear controllers often fail to capture complex interactions. Among these techniques, MPC is one of the most widely studied [10], [26]–[28]. MPC employs the system model to predict future state trajectories and determine the optimal control action within a defined horizon. However, its application to DC–DC boost converters is challenging due to their NMP nature and the computational load of real-time optimization. Long prediction horizons are often required for accurate voltage regulation, increasing processing demands and limiting real-time feasibility [29]. Although move-blocking techniques [30], [31] can reduce this burden but still require powerful processors. Alternatively, regulating both inductor current and output voltage within MPC enables shorter prediction intervals and improves feasibility [32], [33]. Adjusting weighting factors balances voltage and current regulation; however, these short-horizon MPC schemes typically require an additional current sensor, increasing implementation complexity and reducing fault tolerance.

This paper introduces a model-based bang-bang, MBB, controller for DC–DC boost converters that leverages the converter's mathematical model to compensate for the one-sample delay inherent in digital implementation. The proposed method combines the simplicity and fast dynamic response of conventional BB control with a model-based compensation mechanism that updates the state variable in advance, enabling switching decision based on more accurate and timely information. To eliminate the steady-state error in current regulation often observed in BB control, a corrective current term is incorporated into the switching function. Furthermore, to improve robustness and eliminate the need for an additional current sensor—commonly required in recent model-based approaches [32]—an output-current estimation technique is developed. This estimation method enhances the practicality of the proposed strategy and ensures stable converter operation under varying load conditions.

The main contributions of this paper can be summarized as follows:

- Incorporation of inductor-current feedback into the voltage-control law to stabilize short-horizon controllers.
- Integration of a model-based delay compensation in BB control to reduce current ripple while preserving low computational cost and real-time feasibility.
- Introduction of a corrective term in the switching function for voltage and current regulation, effectively removing the steady-state error associated with BB control.
- Development of an output-current estimation method that reduces load dependence and removes the need for an additional sensor, enhancing robustness and practicality.
- Demonstration that the proposed MBB controller closely matches the performance of the short-horizon MPC method introduced in [32], [34], achieving comparable transient response to reference and load variations while retaining the simplicity and low computational cost of conventional BB control.

II. System Description

To enable the proposed compensation mechanism, it is essential to derive an accurate dynamic model that captures the behavior of the DC–DC boost converter under different switching conditions. According to the scheme of the converter from Fig. 1, v_s and v_{out} denote the input and output voltages, and i_L, i_{Load} and i_D represent the inductor current, the load current, and the diode current, respectively. The symbols L, C, S, and D correspond to the inductor, capacitor, switch, and diode. Based on the converter's operation during the ON and OFF switching intervals, the continuous-time dynamic model can be expressed as follows

$$\frac{di_L}{dt} = \frac{1}{L}v_s \tag{1a}$$
$$\frac{dv_{out}}{dt} = \frac{-1}{C}i_{Load} \tag{1b}$$

$$\frac{di_L}{dt} = \frac{1}{L}(v_s - v_{out}) \tag{2a}$$
$$\frac{dv_{out}}{dt} = \frac{1}{C}(i_L - i_{Load}) \tag{2b}$$

In (1), as illustrated in Fig. 3(a), the switch is ON and energy is stored in the inductor, causing i_L to increase. During this mode, the capacitor supplies energy to the load, resulting in a decrease in v_{out}. In (2), as depicted in Fig. 3(b), the switch is OFF and the energy stored in the inductor and the

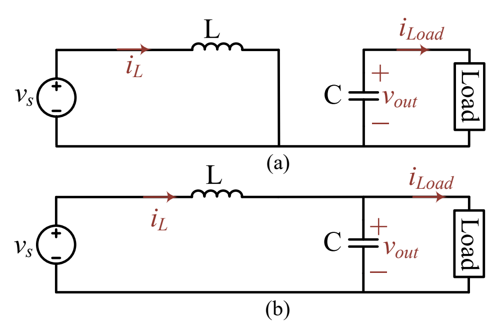


Fig. 3. Converter operating modes: (a) ON switching state and (b) OFF switching state.

input energy are transferred to the output. Consequently, the capacitor charges, causing v_{out} to rise.

III. Application of Bang-Bang Control for Boost Converter

For regulating the boost converter, the BB controller determines the switch state S based on the measured inductor current and output voltage. Depending on whether the control target is i_L or v_{out}, the general form of the switching function is expressed as

$$S = 0.5 \times \Big(1 + \text{sign}\big[f^* - f(k)\big]\Big) \tag{3}$$

where f^* and $f(k)$ denote the reference and measured variables at sampling instant k, respectively. When $f(k) < f^*$, the switch turns ON; otherwise, it turns OFF.

A. Current Regulation

In grid-connected DC microgrids, where the DC bus voltage is maintained by the main grid, converter control primarily focuses on inductor current regulation. The switching function in this case becomes

$$S = 0.5 \times \Big(1 + \text{sign}\big[i_L^* - i_L(k)\big]\Big) \tag{4}$$

where i_L^* is the reference inductor current. The BB controller keeps i_L within a narrow band around i_L^* by toggling the switch ON when the current drops below the setpoint and OFF when it exceeds it.

Fig. 4 illustrates the inductor current behavior under a one-sample digital delay. The reference current steps from 5 A to 15 A at 0.02 s, then to 10 A at 0.04 s. The delay causes larger current ripples and steady-state deviation because switching reacts to outdated measurements. At low duty cycles, the inductor fails to reach the reference value, while at high duty cycles, the current overshoots—demonstrating sensitivity of BB control to sampling delay.

B. Voltage Regulation

In islanded DC microgrids, converter control must ensure accurate DC-bus voltage regulation. If the control law considers only v_{out}, the BB switching function simplifies to

$$S = 0.5 \times \Big(1 + \text{sign}\Big[\big(v_{out}^* - v_{out}(k)\big)\Big]\Big) \tag{5}$$

However, this approach neglects the converter's internal dynamic—particularly i_L—and leads to instability due to the boost converter's NMP behavior. As shown in Fig. 2(b), omitting current regulation causes the system to become unstable, with the output voltage diverging from its reference. This result highlights the inherent limitation of voltage-only switching in BB control when applied to NMP boost converters.

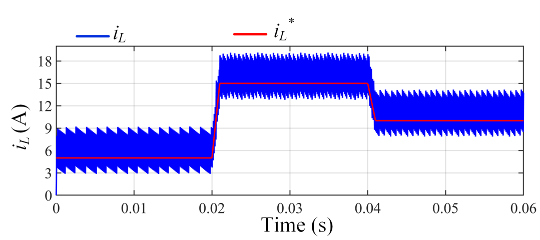


Fig. 4. Current regulation performance of the BB method under one-sample digital delay.

IV. Proposed Model-Based Bang-Bang Control

A. Principle of Proposed Control

To ensure stable operation and limit current ripple, the control objective must consider both output voltage and inductor current. Accordingly, the switching function is modified as shown in Eq. (6) to reflect this dual-objective regulation.

$$S = 0.5\times\Big(1+\text{sign}\Big[\big(v_{out}^*-v_{out}(k)\big)+\lambda\big(i_L^{des}-i_L(k)\big)\Big]\Big) \tag{6}$$

where v_{out}^* and i_L^{des} are the desired output voltage and inductor current, respectively. $v_{out}(k)$ and $i_L(k)$ are the actual values of the output voltage and inductor current at instant k, respectively. λ is a weighting factor used to balance between the objectives of the switching function. It is noted that i_L^{des} is calculated based on power balance, neglecting converter losses as

$$P_{in} \simeq P_{out} \implies i_L^{des} \simeq \frac{v_{out}^* \times i_{Load}}{v_s} \tag{7}$$

where P_{in} and P_{out} are the input and output power of the converter. Based on (7), the load current's information is needed to obtain i_L^{des}. Long-horizon MPC methods reported in the literature [16], [29]–[31] often assume resistive load characteristics, where $i_{Load} = v_{out}/R$. These approaches are therefore load-dependent, requiring model modification when load type changes. In short-horizon MPC methods [32], [33], the load current is measured using an additional current sensor. To eliminate this dependency, the proposed MBB method incorporates a load-current estimation scheme, described in Section IV-B, which allows operation independent of load type and eliminates the need for an additional load current sensor by estimating the load current, while the inductor current (a required state variable for system dynamics) is still measured.

The proposed switching function in its fundamental form (6) was implemented in a simulation environment to evaluate the effectiveness of the BB controller in voltage regulation, while accounting for the one-sample delay introduced by digital implementation (see Fig. 5). As shown in Fig. 4 and discussed in Section III-A, the bang-bang controller exhibits increased current ripple and a steady-state error when subject to digital delay. To reduce the current ripple, the proposed MBB control strategy incorporates the converter's discrete-time model to update the inductor, as described in Section IV-C. Moreover, to mitigate the steady-state current error, a corrective term is added to the switching function, as explained in Section IV-D.

B. Output Current Estimation Method

To achieve fully model-based operation without requiring an additional current sensor, the output current is estimated using Kirchhoff's current law at the converter output node as

$$i_{Load}(k) = i_D(k) - C\frac{v_{out}(k) - v_{out}(k-1)}{T_s} \tag{8}$$

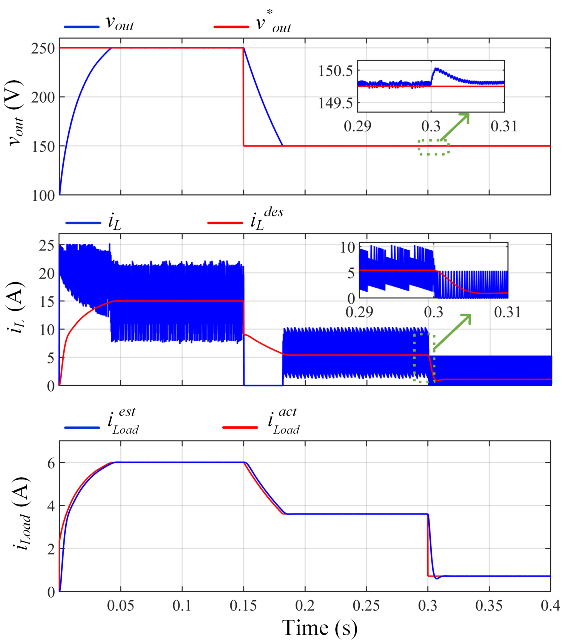


Fig. 5. Voltage regulation performance of the conventional BB method under one-sample digital delay.

Here, $i_D(k)$ represents the diode current at the current sampling instant, which is obtained by

$$i_D(k) = \big(1 - S(k-1)\big) \times i_L(k) \tag{9}$$

Equation (9) accounts for the one-sample delay, necessitating the use of the switching state from the previous sampling instant ($S(k-1)$) to accurately estimate the output current. As a result, the final expression for the output current estimation is derived by improving the accuracy of the current estimation through the use of the average inductor current over two consecutive sampling periods given by

$$\begin{aligned} i^{est}_{Load}(k) = \big(1 - S(k-1)\big) \times \frac{i_L(k) + i_L(k-1)}{2} \\ - C\frac{v_{out}(k) - v_{out}(k-1)}{T_s} \end{aligned} \tag{10}$$

To further refine the output current estimation process, the potential measurement noise from the voltage sensor is considered. To this end, a second-order low-pass filter is designed to minimize the impact of this noise, ensuring a more precise estimation. The filter's characteristic is given by

$$T(s) = \frac{K\omega_c^2}{s^2 + 2\zeta\omega_c s + \omega_c^2} \tag{11}$$

where K denotes the filter gain (set to unity), $\omega_c = 2\pi f_c$ is the cut-off angular frequency, and ζ is the damping ratio.

To provide a systematic basis for selecting the cut-off frequency f_c, the filter dynamics can be analyzed through its pole locations:

$$s_{1,2} = -\zeta\omega_c \pm j\omega_c\sqrt{1-\zeta^2}. \tag{12}$$

These poles are directly determined by ω_c. Increasing ω_c shifts the poles further into the left-half plane, resulting in faster response but reduced attenuation of high-frequency noise. Conversely, decreasing ω_c improves noise suppression by attenuating high-frequency components, but introduces additional phase delay and limits the ability of the estimator to track fast signal variations. Therefore, the selection of ω_c involves a trade-off between noise attenuation and dynamic performance.

In this application, the dominant disturbance originates from switching harmonics at the switching frequency f_s and its multiples. Accordingly, the cut-off frequency should be selected sufficiently lower than f_s to attenuate switching-induced ripple, while remaining higher than the dominant low-frequency dynamics governed by the output LC time constants. Based on this frequency separation principle [35], the cut-off frequency is selected to satisfy this trade-off.

For the system under consideration with $f_s = 50\,\text{kHz}$, a cut-off frequency of $f_c = 200\,\text{Hz}$ is selected as an effective compromise between noise attenuation and dynamic performance. The damping ratio ζ is fixed at 0.707 to achieve a well-damped response with minimal overshoot. To enable digital implementation, the discrete form of the filter is derived using the backward Euler method with a sampling frequency of 50 kHz, as follows:

$$T(z) = \frac{0.0006316547}{1.0361693508z^{-2} - 2.0355376961z^{-1} + 1} \tag{13}$$

It should be noted that the proposed low-pass filter is applied only within the load current estimation loop and does not directly modify the primary control law. Due to the significant separation between the filter bandwidth and the converter switching frequency, the phase delay introduced by the filter remains negligible. As a result, the transient response is effectively preserved, with no observable degradation in voltage dynamics under load step conditions.

Furthermore, the inclusion of the filter is essential in practical implementations to suppress unavoidable measurement noise, thereby ensuring reliable estimation and robust control performance.

C. Model-Based Current Update

To realize the model-based compensation mechanism, the dynamic equations of the boost converter current are discretized to update the inductor current one sampling step ahead. Using the Euler method applied to the continuous-time models (1a) and (2a), the discrete forms are obtained as

$$i_L(k+1) = \frac{T_s}{L}v_s(k) + i_L(k) \tag{14a}$$

$$i_L(k+1) = \frac{T_s}{L}(v_s(k) - v_{out}(k)) + i_L(k) \tag{14b}$$

where T_s is the sampling period of the controller. Depending on the switching state selected in the previous computational cycle and realized in the cycle k, either (14a) or (14b) is used to compute the updated inductor current. The compensated current is then applied in the BB control law to determine the switching action for time step $k+1$. As demonstrated later, this approach effectively reduces current ripple.

Although updating the DC bus voltage signal is also possible, it can be omitted because of the large capacitance typically used in the DC link, which results in negligible voltage variation within a single switching period. Consequently, the one-sample delay introduced by digital implementation has an insignificant influence on the voltage dynamics. Therefore, unlike conventional model-based predictive control strategies that update both current and voltage at each step, the proposed method excludes voltage updating. This omission avoids redundant computations, reduces the overall computational burden, and enhances real-time feasibility without compromising control performance.

D. Modified Switching Function

The compensated current variable calculated from (14) is used in the BB switching function to determine the next control action in both, current regulation and voltage regulation modes.

1) Current Regulation: As illustrated in Fig. 4, the conventional BB controller can introduce steady-state error at certain duty cycles. To address this issue, a corrective term is incorporated into the current regulation switching function. This enhancement allows the controller to more accurately adjust the inductor current, thereby eliminating steady-state error and improving current tracking precision. This enhancement is particularly critical when operating at duty cycles significantly lower or higher than 0.5, where the inductor may either lack sufficient time to build up the required current or risk saturation. The modified switching function for current regulation is expressed as

$$S = 0.5 \times \Big(1 + \text{sign}\big[(i_L^{des} + i_\Delta) - i_L(k+1)\big]\Big) \tag{15}$$

where i_Δ is the corrective term added to the switching function. This term, which represents the average change in inductor current during the ON and OFF states, is given by

$$i_\Delta = 0.5 \times \big(2v_s(k) - v_{out}(k)\big)\frac{T_s}{L} \tag{16}$$

It should be noted that the corrective term i_Δ is introduced to compensate for the average inductor current variation over a switching cycle, thereby eliminating steady-state error and improving current tracking accuracy.

The switching expression $(i_L^{des} - i_L(k+1) + i_\Delta)$ can be interpreted as the combination of a current tracking error term and a feedforward corrective component. The current error term governs the dynamic behavior of the system, while i_Δ shifts the effective current reference to compensate for steady-state offset caused by switching asymmetry. Importantly, i_Δ does not introduce additional system dynamics, as it does not contain derivative or state-dependent dynamic terms. Therefore, it does not introduce new poles into the closed-loop system and does not affect the stability mechanism. Consequently, the closed-loop stability is primarily governed by the switching function structure, while i_Δ enhances steady-state accuracy.

The performance of the proposed method in current regulation is illustrated in Fig. 6. As observed, the inductor current is accurately regulated, exhibiting negligible steady-state error and reduced ripple compared with the BB controller (Fig. 4). In this figure, i_L^{comp} represents the compensated inductor current, which corresponds to the update current value at the next sampling interval. As demonstrated, the compensated current at instant k matches the measured current at instant $k+1$, ensuring that the switching decision is based on the correct current information. Although pure current regulation using the BB controller is not the primary objective of this work, the corrective action introduced through i_Δ will be employed in voltage regulation, where the control law incorporates both output-voltage and inductor-current terms.

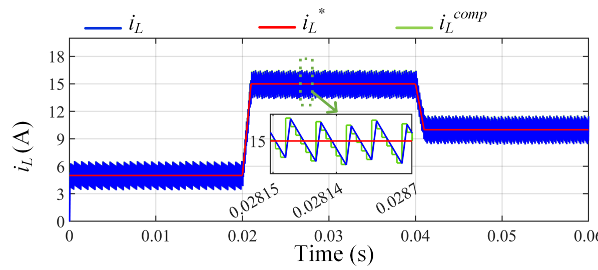


Fig. 6. Current regulation performance of the proposed MBB control method.

2) Voltage Regulation: In voltage-regulated operation, the objective of the converter is to maintain the output voltage v_{out} at its reference value v_{out}^* despite variations in load or source voltage. In the proposed MBB controller, only the inductor current i_L is compensated—using the model-based to update $(i_L(k+1))$ together with a corrective current term i_Δ —while the measured output voltage v_{out} is used directly in the switching function. The resulting switching function is defined as

$$\begin{aligned} S = 0.5 \times \Big(1 + \text{sign}\Big[&\big(v_{out}^* - v_{out}(k)\big) + \\ &\lambda\big((i_L^{des} + i_\Delta) - i_L(k+1)\big)\Big]\Big) \end{aligned} \tag{17}$$

The first term $v_{out}^* - v_{out}(k)$ drives the output voltage toward its reference, while the second term provides a current-based correction that accelerates transient recovery and improves stability under non-minimum-phase conditions. Consistent with the previous discussion, the corrective term i_Δ acts as a feedforward component that adjusts the reference level of the current, while the stability and dynamic response are primarily determined by the feedback terms associated with the voltage and current errors in the switching function. Furthermore, the use of the compensated inductor current $i_L(k+1)$ ensures that the switching action is based on the predicted current at the next sampling instant, thereby mitigating the effects of digital delay and improving dynamic response.

The weighting factor λ determines the relative contribution of the voltage and current errors in the switching decision and plays a key role in shaping the dynamic response of the controller. In particular, λ governs the trade-off between voltage regulation and current error. A small value of λ reduces the influence of current feedback, which may degrade damping and stability due to the NMP behavior of the converter. In contrast, a large value of λ increases the dominance of the current term, which improves current regulation but may slow down the voltage response and can introduce steady-state deviation in the output voltage.

In this work, λ is tuned using an offline optimization procedure to achieve an appropriate balance between fast voltage tracking, reduced settling time, and acceptable current ripple. The obtained results indicate that λ is primarily influenced by the output capacitance C, which governs the voltage dynamics of the converter. Larger capacitance values provide stronger energy buffering and inherently smoother voltage behavior, thereby reducing the need for aggressive current correction and leading to smaller optimal values of λ. Conversely, smaller capacitance values require larger λ to ensure stable and well-damped response. In contrast, the inductance L mainly affects the current dynamics, which are directly regulated within the control loop, and therefore has a less significant impact on the weighting factor. Based on the optimization results, an empirical tuning guideline is obtained as

$$\lambda(C) = 7.48\, C(\mu F)^{-0.547}$$

which serves as a practical design rule within the considered operating range.

The complete structure of the proposed MBB voltage controller is illustrated in Fig. 7. After computing the switching signal, an over-current protection condition is imposed: if the compensated inductor current, $i_L(k{+}1)$, exceeds its maximum allowable value, i_L^{max}. If the compensated current exceeds i_L^{max}, the switch is deactivated; otherwise, the switching state derived from the modified switching function is applied.

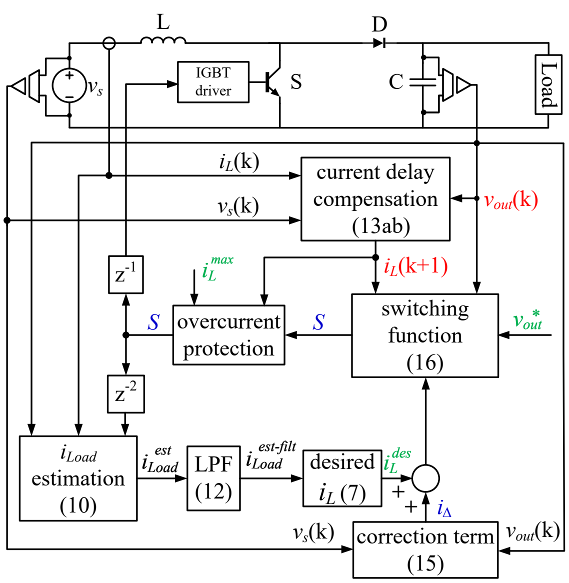


Fig. 7. The block diagram of the proposed MBB control method for voltage regulation.

V. Stability Analysis of the Proposed MBB Control Method

The proposed MBB control strategy incorporates both output voltage and inductor current feedback to address the NMP behavior of the boost converter. The switching function, previously defined in (17), is based on a composite error that combines voltage tracking and current regulation, where the voltage and current tracking errors are defined as

$$e_v(k) = v_{out}^* - v_{out}(k) \tag{18}$$

$$e_i(k) = (i_L^{des} + i_\Delta) - i_L(k+1). \tag{19}$$

The composite switching surface is defined as

$$\sigma(k) = e_v(k) + \lambda e_i(k). \tag{20}$$

Due to the switching nature of the controller and the presence of nonidealities such as parameter uncertainty, estimation errors, and measurement noise, an exact closed-form expression of the error dynamics is not analytically tractable. However, the closed-loop behavior can be characterized in a bounded form as

$$|\sigma(k+1)| \le \rho|\sigma(k)| + |d(k)| \tag{21}$$

where $0 < \rho < 1$ represents the contraction effect induced by the control action and $d(k)$ is a bounded disturbance term.

To analyze stability, consider the discrete-time Lyapunov candidate function

$$V(k) = \frac{1}{2}\sigma^2(k). \tag{22}$$

The corresponding Lyapunov difference satisfies

$$\Delta V(k) = V(k+1) - V(k) \le -\alpha V(k) + \beta \tag{23}$$

where $\alpha > 0$ and $\beta > 0$ are constants determined by system parameters and disturbance bounds. This inequality guarantees that the closed-loop system is uniformly ultimately bounded (UUB). Therefore, all system states remain bounded and converge to a finite neighborhood of the equilibrium point despite the presence of disturbances and modeling uncertainties.

From a physical perspective, the proposed MBB control stabilizes the internal energy dynamics of the converter by regulating the inductor current. This effectively mitigates the instability mechanism associated with the right-half-plane zero of the boost converter and ensures stable voltage regulation.

VI. Results and Discussions

The effectiveness of the proposed strategy is evaluated through simulation studies, comparative analysis with other control methods, and experimental validation. The obtained results are benchmarked against those of the BB control, as discussed in previous sections. Table I summarizes the system parameters used in both simulations and experiments.

TABLE I
System Parameters

Parameter	Symbol	Value
Input Voltage	v_s	100 V
Output Voltage Range	v_{out}^*	150 V – 250 V
Maximum Inductor Current Value	i_L^{max}	20 A
Resistive Load Range	R	41.6 Ω – 416 Ω
Constant Current Load	i_{CCL}	6 A
Constant Power Load	P_{CPL}	1.5 kW
Input Inductance	L	750 μH
Output Capacitance	C	1500 μF
Sampling Time	T_s	20 μs

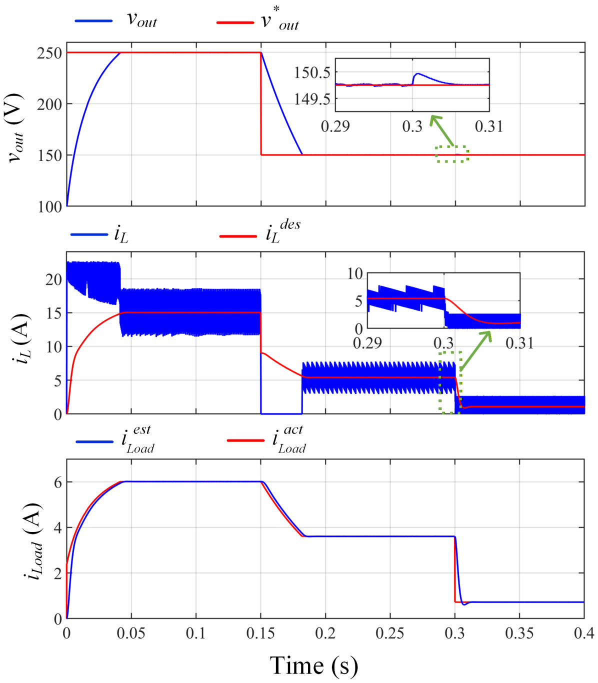


Fig. 8. Voltage regulation performance of the proposed MBB control method.

A. Simulation Results

1) Proposed BB Control Performance: The performance of the proposed MBB controller in voltage regulation is evaluated under three operating conditions, as shown in Fig. 8. The reference voltage is initially set to 250 V at $t = 0$ s and reduced to 150 V at $t = 0.15$ s. To evaluate the performance of the proposed strategy in both continuous-conduction mode (CCM) and during the transition to discontinuous-conduction mode (DCM), the load resistance is varied from 41.6 Ω to 208 Ω at $t = 0.3$ s after the system stabilizes. As observed, the proposed controller precisely regulates the output voltage, significantly reduces inductor-current ripple, and mitigates steady-state error through the corrective current term. Moreover, the estimated output current closely follows the measured value, confirming the accuracy and effectiveness of the proposed estimation method.

2) Parameter Mismatch Robustness: To evaluate the robustness of the proposed MBB controller, its performance is first examined under moderate parameter variations, specifically a 20% reduction in both the output capacitance and inductance of the plant. As shown in Fig. 9, this variation has a negligible impact on voltage regulation and overall system stability. The controller maintains accurate steady-state performance, with no noticeable deviation in output voltage or current ripple. A slightly faster transient response is observed, which is attributed to the reduced energy storage of the passive components rather than any limitation of the control strategy.

To further investigate robustness limits, more severe mismatch conditions are considered by applying simultaneous $\pm 50\%$ variations in both the inductance (L) and capacitance (C) of the plant. These extreme deviations represent conservative worst-case scenarios that exceed typical component tolerances and aging effects. The simulation results for $+50\%$ and -50% variations are depicted in Fig. 10 and Fig. 11, respectively. As can be seen, the output voltage continues to closely track the reference value with no steady-state error under all conditions, confirming the strong robustness of the proposed MBB controller.

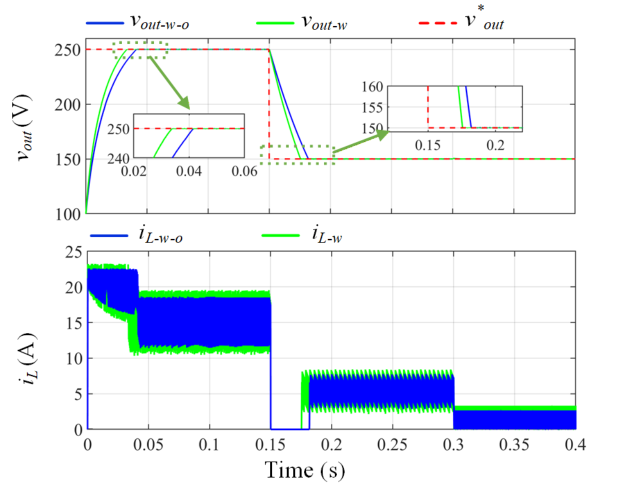


Fig. 9. Voltage regulation performance of the proposed MBB control method under a 20% parameter mismatch. The subscripts "w" and "w-o" denote the cases with and without parameter mismatch, respectively.

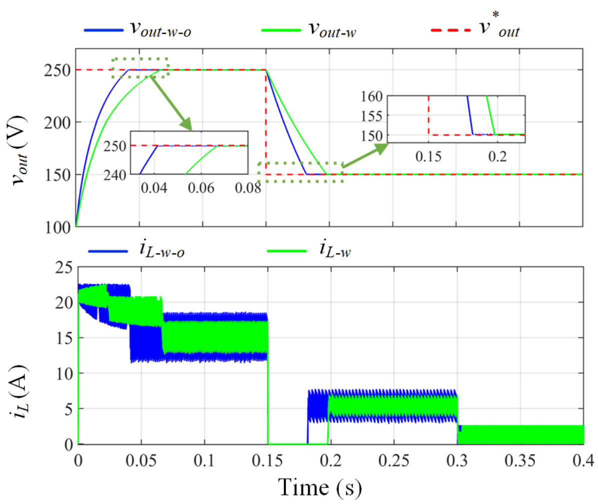


Fig. 10. Voltage regulation performance of the proposed MBB control method under a 50% increase in parameter mismatch. The subscripts "w" and "w-o" denote the cases with and without parameter mismatch, respectively.

It is worth noting that the observed variations in system behavior are primarily governed by the physical characteristics of the passive components. Specifically, a decrease in inductance increases the inductor current ripple, while a higher inductance results in smoother current waveforms. Similarly, a lower capacitance leads to faster voltage dynamics, whereas a higher capacitance produces a slower but smoother transient response. Despite these variations, the controller consistently maintains stable operation, accurate voltage regulation, and bounded current behavior. No instability or degradation in control performance is observed across all tested scenarios.

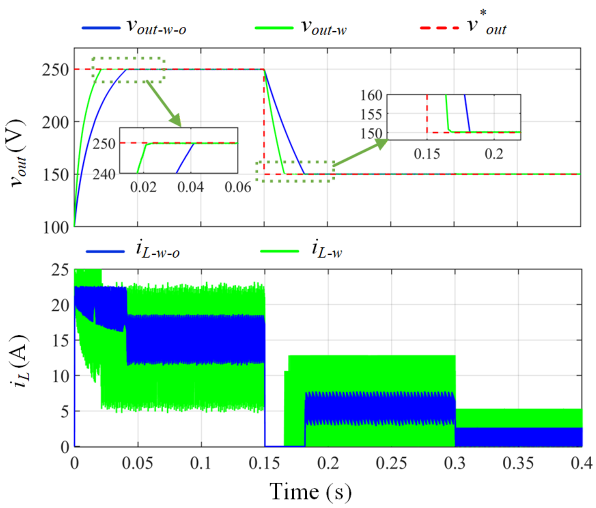


Fig. 11. Voltage regulation performance of the proposed MBB control method under a 50% decrease in parameter mismatch. The subscripts "w" and "w-o" denote the cases with and without parameter mismatch, respectively.

B. Comparative Evaluation

To further validate the effectiveness of the proposed MBB controller, comparative simulations were conducted in two stages. In the first stage, the controller was compared with two widely used voltage-control strategies for boost converters—PI–MPC and PI–BB—where a PI regulator forms the outer voltage loop, and MPC or BB governs the inner current loop. In the second stage, the proposed method was compared with a recently reported short-horizon MPC (SH-MPC) [34], which simultaneously regulates voltage and current through a single predictive framework.

1) Comparison with Cascaded Controllers: As shown in Fig. 12, all controllers successfully regulate the output voltage; however, their transient behaviors differ significantly. The proposed MBB controller achieves the fastest response and exhibits negligible overshoot or undershoot during reference voltage transitions at 0.15 s and 0.3 s. In contrast, the PI–MPC and PI–BB strategies show slower settling and larger transient deviations.

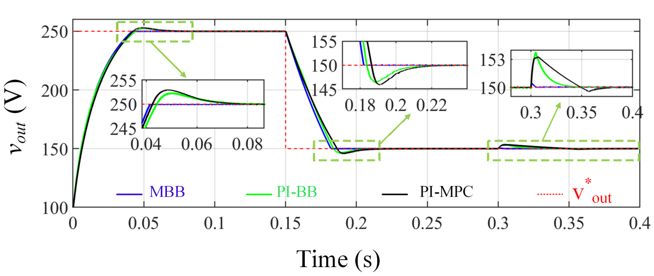


Fig. 12. Voltage regulation comparison of PI–MPC, PI–BB, and proposed MBB methods.

2) Comparison with Short-Horizon MPC: In the second stage, the proposed MBB control strategy is compared with the SH-MPC method [34] under identical test conditions (Fig. 13). The results show that both methods exhibit nearly identical dynamic and steady-state performance, confirming that the proposed model-based compensation successfully reproduces the predictive behavior of SH-MPC. However, the MBB controller demonstrates lower voltage ripples compared with SH-MPC, highlighting its superior steady-state smoothness.

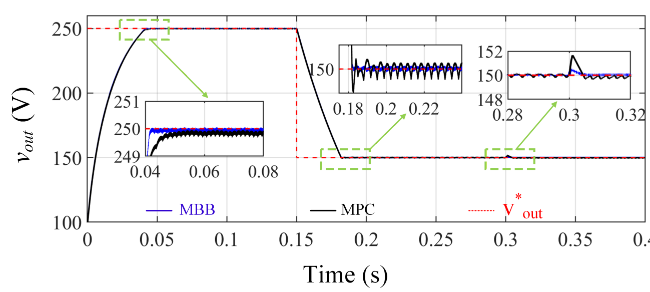


Fig. 13. Voltage regulation comparison of short-horizon MPC [34] and proposed MBB methods.

Moreover, unlike SH-MPC—which requires solving an optimization problem at each sampling instant and updating both current and voltage states—the MBB approach determines the switching state through an explicit control law. By omitting the voltage updating step, computational burden is further reduced without affecting accuracy. This simplicity enables real-time implementation on low-cost digital controllers while maintaining fast transient response and high control precision. To quantitatively support this claim, a comparison of the computational requirements imposed by both strategies on the target DSP is presented in Table II.

TABLE II
COMPUTATIONAL COMPLEXITY COMPARISON BETWEEN PROPOSED MBB AND SHORT-HORIZON MPC [34]

Metric	SH-MPC	MBB
Multiplications	16	8
Add./Sub.	16	8
Comparisons	≈2 (cost+dec.)	1 (sign)
Steps	$k+1, k+2$	$k+1$
Cost eval.	Required (per state)	Not required
Clock cycles	≈34	≈17
Exec. time (ns)	≈227	≈113

As observed, the estimated execution time of the proposed MBB controller is approximately 113 ns, whereas the MPC method requires approximately 227 ns. This indicates that the proposed method reduces the computational burden by nearly 50%, primarily due to the elimination of prediction step and cost function evaluation.

3) Quantitative Comparison: To improve the clarity and readability of the comparison, a zoomed-in view focusing on the load step interval is presented in Fig. 14. This representation brings together the responses of all control methods within the same time window, allowing a direct and fair comparison of their transient behavior. By concentrating on this critical operating region, the differences in dynamic performance—such as overshoot, settling time, and transient current response—are more clearly observable.

To further quantify these differences, key performance metrics are extracted from the load step response and summarized in Table III. The selected metrics include voltage settling time, voltage overshoot, and inductor current ripple, providing a

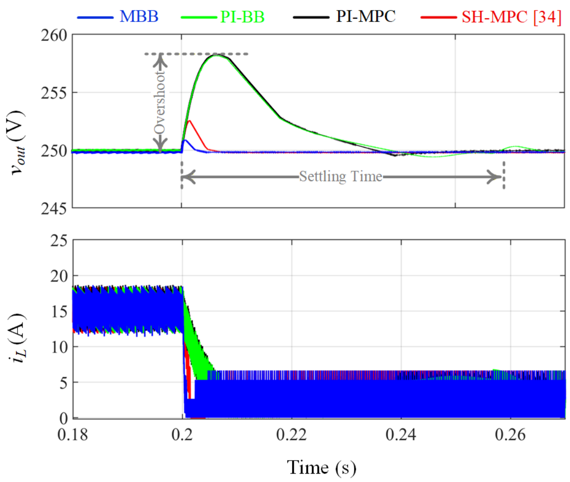


Fig. 14. Voltage regulation comparison of the proposed MBB control method with PI-BB, PI-MPC, and short-horizon MPC [34] under a load step.

comprehensive assessment of transient performance. As can be seen, the proposed MBB control method achieves the fastest settling time and the lowest overshoot among the considered approaches, while maintaining a comparable level of current ripple. This demonstrates that the proposed method provides an improved trade-off between dynamic performance and implementation simplicity.

TABLE III
QUANTITATIVE COMPARISON UNDER LOAD STEP CONDITIONS

Method	Settling time (ms)	Volt. Over-shoot (V)	Current Ripple (A)
PI-BB	53.13	8.15	5.5
PI-MPC	43.75	8.25	5.8
SH-MPC [34]	7.81	2.5	6.0
MBB	4.68	0.8	6.0

C. Experimental Results

The effectiveness of the proposed MBB method is validated via experimental results. The experimental setup depicted in Fig. 15 is used for this purpose, utilizing a Texas Instrument TMS320F28335 microcontroller operating at a clock cycle of 150 MHz with 12-bits ADC channels. The results are obtained under varying load conditions, including resistive, constant current (CC), and constant power (CP) loads, to demonstrate the robustness of the proposed method.

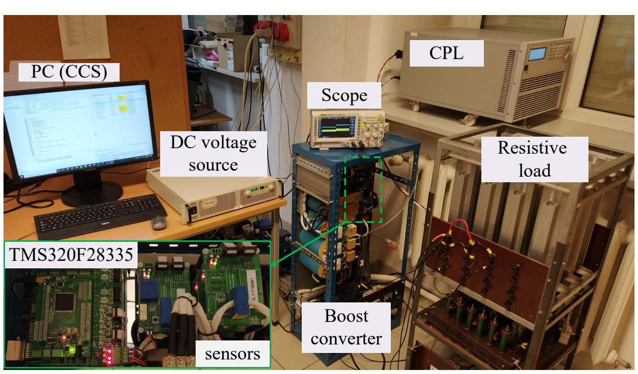


Fig. 15. Testbed for experimental validation of the proposed method

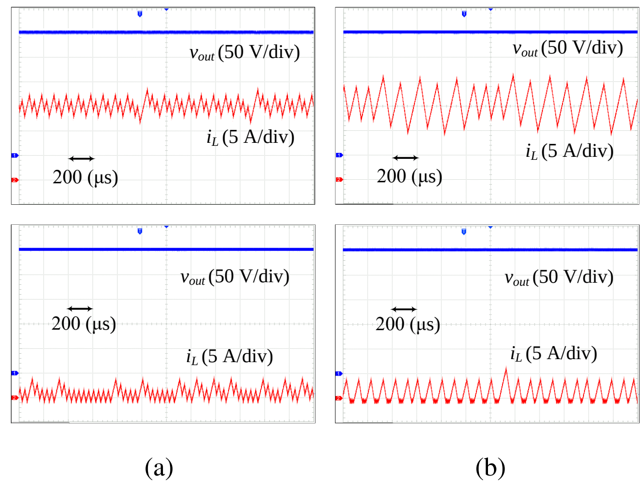


Fig. 16. Steady-state behavior of the boost converter with a 41.6 Ω resistive load in CCM (first row) and a 416 Ω resistive load in DCM (second row): (a) proposed MBB control and (b) BB control for a reference voltage of 250 V.

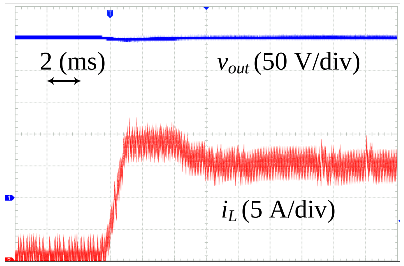
1) Case 1 – Resistive Load: The proposed controller is experimentally validated using resistive loads under both steady-state and transient conditions. To assess its performance across CCM and DCM, two loads are tested: 41.6 Ω and 416 Ω.

Fig. 16 shows the steady-state waveforms at a 250 V reference. The first row corresponds to CCM operation and the second to DCM. In each case, the left column presents results for the proposed MBB method, while the right shows those of the BB controller. As observed, both methods maintain the desired output voltage; however, the MBB controller achieves approximately 50% lower current ripple and eliminates steady-state error in both modes. This improvement results from the combined effect of the model-based current update and the corrective term, which together ensure smoother current transitions and improved voltage stability.

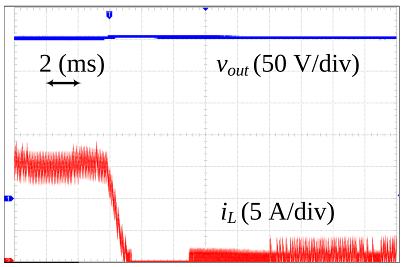
To further evaluate the proposed controller under dynamic conditions, transient tests are performed using resistive loads. In the first scenario, the load is changed from 416 Ω to 41.6 Ω and vice versa, with the reference voltage fixed at 250 V (Fig. 17). In the second scenario, the load is held constant at 41.6 Ω while the reference voltage is varied—first increasing from 150 V to 250 V, then decreasing back to 150 V (Fig. 18).

As shown, the proposed MBB controller provides accurate voltage tracking and rapid recovery following both load and reference variations. Compared with the BB control, it achieves lower current ripple, faster settling, and no observable overshoot or undershoot. These improvements are caused by updating the inductor current, which compensates for digital delay while preserving the fast response of BB control.

2) Case 2 – Non-resistive Loads: The converter's dynamic performance is evaluated under CC and CP loads to verify that the proposed method is independent of load type. The reference voltage is stepped between 150 V and 250 V (both directions), as shown in Fig. 19 and Fig. 20. Tests used a 6 A CC load (1.5 kW at 250 V) and a 1.5 kW CP load at the converter output.

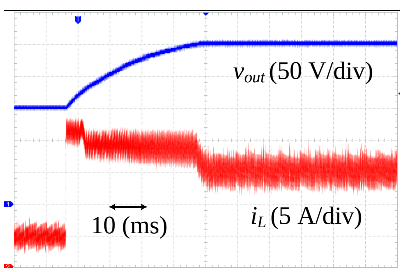


(a)

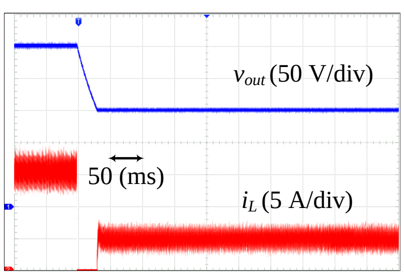


(b)

Fig. 17. Dynamic response of the boost converter using proposed MBB to load changes: (a) from 416 Ω to 41.6 Ω, and (b) from 41.6 Ω to 416 Ω, under reference voltage of 250 V.



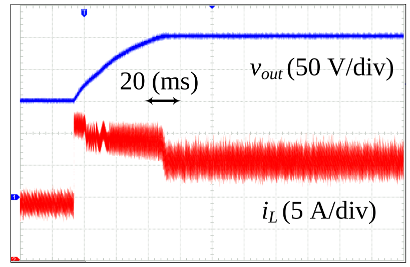
(a)



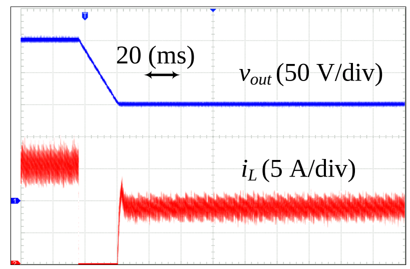
(b)

Fig. 18. Dynamic response of the boost converter using the proposed MBB control during reference voltage transitions under a 41.6 Ω resistive load: (a) step-up from 150 V to 250 V, and (b) step-down from 250 V to 150 V.

As observed, regardless of the load type, the proposed MBB method accurately tracks the reference voltage while maintaining lower inductor current ripple compared with the BB method, which suffers from high ripple caused by digital implementation. The results also highlight the effectiveness of the output current estimation method, which accurately determines the required inductor current for switching decisions without requiring prior knowledge of the load type or additional current sensor.

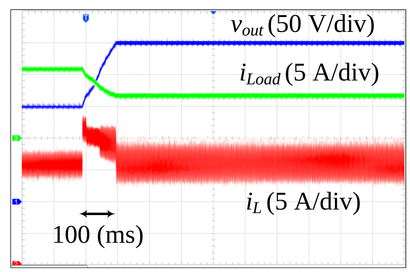


(a)

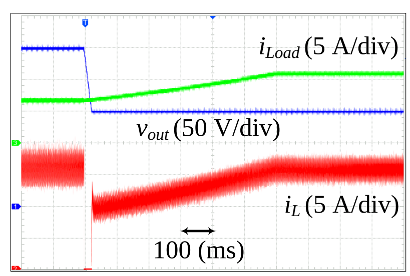


(b)

Fig. 19. Dynamic response of the boost converter using proposed MBB to reference voltage changes from: (a) 250 V to 150 V (b) 150 V to 250 V, under a 6 A constant current load.



(a)



(b)

Fig. 20. Dynamic response of the boost converter using proposed MBB controller to a reference voltage change from: (a) 250 V to 150 V, and (b) from 150 V to 250 V under a 1.5 kW constant power load.

VII. Conclusion

This paper presented a model-based bang-bang control strategy for DC–DC boost converters. While BB control offers simplicity and fast response, it suffers from current ripple and steady-state error due to one-sample digital delay. The proposed MBB method employs a model-based compensation mechanism to update state variables in advance, ensuring accurate switching decisions. A corrective current term eliminates steady-state error, and an output-current estimation technique removes dependence on load characteristics and the need for an extra sensor.

Simulation and experimental results verify that the MBB controller achieves fast and precise voltage regulation with significantly reduced ripple, maintaining stable operation under resistive, constant-current, and constant-power loads. Overall, the MBB strategy combines BB simplicity with model-based state updating, providing a robust and computationally efficient solution for DC–DC converters in modern microgrids.

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